



Question 3: Exam Score Analysis (NumPy)

1. Matrix Creation

```
In [18]: import numpy as np

exam_scores = np.array([
    [85, 92, 78, 90],
    [76, 88, 92, 80],
    [90, 85, 88, 95],
    [82, 78, 88, 75],
    [88, 90, 85, 92]]
)

exam_scores
```

```
Out[18]: array([[85, 92, 78, 90],
               [76, 88, 92, 80],
               [90, 85, 88, 95],
               [82, 78, 88, 75],
               [88, 90, 85, 92]])
```

b. Key Difference between Python List and Numpy Array:

A Python list can store elements of different data types and is slower for numerical operations. A NumPy array stores elements of the same data type and is optimized for fast mathematical and numerical computations.

2. Statistical Analysis

```
In [19]: max_score = np.max(exam_scores, axis=0)
min_score = np.min(exam_scores, axis=0)

print("Maximum score per subject: ", max_score)
print("Minimum score per subject: ", min_score)
```

```
Maximum score per subject: [90 88 92 94]
Minimum score per subject: [76 78 78 75]
```

```
In [20]: class_average = np.mean(exam_scores)
print("Average score of the entire class: ", class_average)
```

```
Average score of the entire class: 85.85
```

3. Advanced Analysis

Find Student (Row Index) who had the highest total score

```
In [21]: total_scores_per_student = np.sum(exam_scores, axis=1)

print(total_scores_per_student)

top_student_idx = np.argmax(total_scores_per_student)

print("Student with highest total score (row index): ", top_student_idx)
```

```
[345 336 358 323 355]
```

```
Student with highest total score (row index): 2
```

Find the Subject (Column Index) that had the lowest average score

```
In [22]: average_score_per_subject = np.mean(exam_scores, axis=0)

print(average_score_per_subject)

lowest_subject_idx = np.argmin(average_score_per_subject)

print("Subject with lowest average score (column index): ", lowest_subject_idx)
```

```
[84.2 86.6 86.2 86.4]
```

```
Subject with lowest average score (column index): 0
```